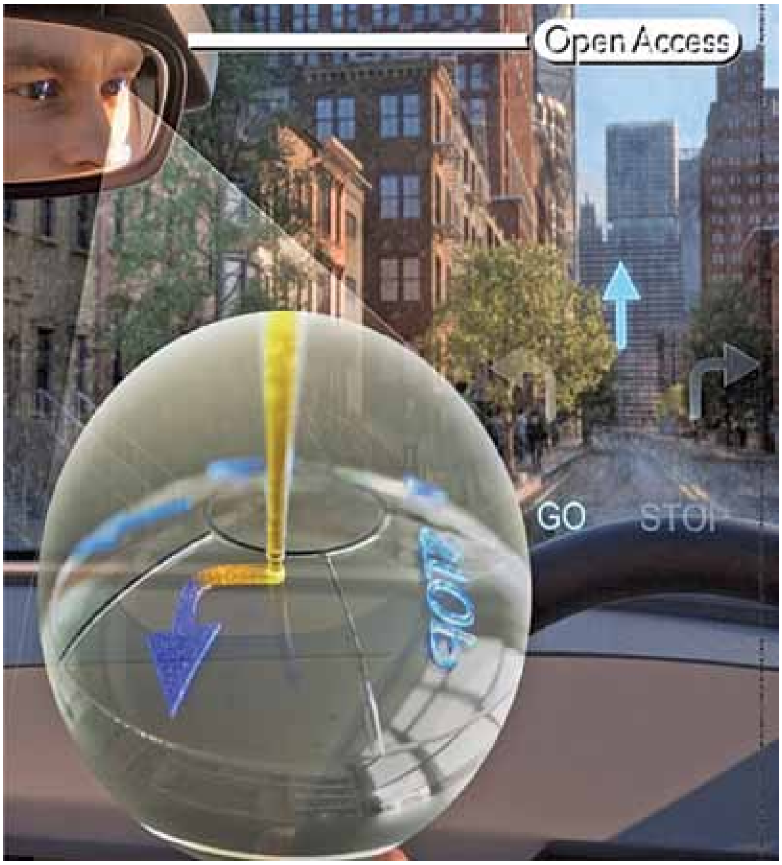
Smart contact lenses enable AR based navigation

A team of researchers from Ulsan National Institute of Science and Technology (UNIST) and KERI have developed core technology for smart contact lenses that can implement augmented reality (AR) based navigation with a 3D printing process. The key is the meniscus of used ink. The meniscus of the acidic-ferric-ferricyanide ink is formed on the substrate when the ink-filled micronozzle and substrate come in contact. Through the precise movement of the nozzle, the crystallisation of Prussian Blue is continuously performed, thereby forming micro-patterns. Simply by wearing a lens, navigation unfolds in front of a person's eyes through AR. Games such as the popular 'Pokemon Go' can also be enjoyed with the smart contact lenses. This technology is expected to be used mainly for navigation but can also be adapted for several other applications.



Meniscus-guided micro-printing of Prussian Blue is realised by the localised crystallisation of FeFe(CN)6 on the substrate confined by the ink meniscus and thermal reduction of the crystallised FeFe(CN)6. This strategy is capable of being used as an electrochromic display to give real-time directions on an augmented reality (AR) smart contact lens device (Credit: Advanced Science and Korea Electrotechnology Research Institute)

A new way to govern ethical use of AI

Researchers from Texas A&M University School of Public Health have developed a new governance model for ethical guidance and enforcement in the rapidly advancing field of artificial intelligence (AI) known as Copyleft AI with Trusted Enforcement, or CAITE. This model combines aspects of copyleft licensing and the patent-troll model—the two methods of managing intellectual property rights that can be considered at odds with one another. It uses a copyleft approach to ensure that developers who create derivative models and data must also use the same license terms as the parent works. The license would assign the enforcement rights of the license to a designated third-party known as a CAITE host. Hosts can set consequences for unethical actions, such as financial penalties or reporting instances of consumer protection law violations. The CAITE approach allows for leniency policies that can promote self-reporting and gives flexibility that typical government enforcement schemes often lack.